CyberHX - Kindle Reader ("Think Like a Hacker")

`CyberHX{y0u_d0nt_l34rn_l1nux_y0u_t4k3_c0ntr0l}`

1. Recon the home folder

Extracted `reader_home.tar.gz`. `.bash_history` is a decoy (`history -c`), but `.bashrc` redirects `HISTFILE` to `~/.local/state/.h`. There's also a hidden `~/.../q` dir and the Kindle clippings copied to `~/.local/share/.mc`.

```
…/Pinaka/reader_home ❱ find reader -type f
reader/.../q
reader/.bash_history
reader/.bashrc
reader/.local/share/.mc
reader/.local/state/.h
reader/.motd
reader/.profile
reader/.vault/flag.enc
reader/Documents/reading-list.md
reader/notes/todo.md

…/Pinaka/reader_home ❱
```

Fig 1 - files in the home folder

2. The real history = the key recipe

pick N takes the Nth word (split on anything not A-Za-z or ASCII apostrophe, lowercased). **note i** takes the i-th `w=` from the clippings. The key is `w1-w2-w3-w4-<asin lowercase>`, then AES-256-CBC, PBKDF2, 250000 iterations, SHA-256.

```
…/Pinaka/reader_home ❱ cat ~/.local/state/.h | grep -v "^#"
cd ~
ls -la
mount | grep -i kindle
cp "/media/reader/Kindle/documents/My Clippings.txt" ~/.local/share/.mc
grep -c 'clipping limit' ~/.local/share/.mc
Q=$(rev ~/.../q | base64 -d)
xdg-open "https://www.amazon.in/s?k=${Q// /+}&i=digital-text"
export ASIN=$(cat ~/.cache/.asin)
pick(){ tr -cs "A-Za-z'" '\n' | sed '/^$/d' | sed -n "${1}p" | tr 'A-Z' 'a-z'; }
note(){ grep -o 'w=[0-9]*' ~/.local/share/.mc | sed -n "${1}p" | cut -d= -f2; }
vim /dev/shm/h1 /dev/shm/h2 /dev/shm/h3 /dev/shm/h4
for i in 1 2 3 4; do pick "$(note $i)" < /dev/shm/h$i; done | paste -sd- | tr -d '\n' >
  /dev/shm/.k
printf -- '-%s' "${ASIN,,}" >> /dev/shm/.k
openssl enc -aes-256-cbc -salt -pbkdf2 -iter 250000 -md sha256 -in ~/flag.txt -out
  ~/.vault/flag.enc -pass file:/dev/shm/.k
shred -u ~/flag.txt /dev/shm/h1 /dev/shm/h2 /dev/shm/h3 /dev/shm/h4 /dev/shm/.k
  ~/.cache/.asin
sed -i -E 's/^(\xEF\xBB\xBF)?[^(]+\(K[a-z]+ D[a-z]+\)\r$/\1[REDACTED] (K******* D******)\r/'
  ~/.local/share/.mc
history -w

…/Pinaka/reader_home ❱
```

Fig 2 - ~/.local/state/.h

3. Clippings + search query

The title was sed'd to `[REDACTED] (K******* D******)`, but the notes survived: word index / word count and an md5 prefix for each highlight. q reversed and base64-decoded gives the Kindle store search.

```
…/Pinaka/reader_home ❱ cat ~/.local/share/.mc | tr -d "\r" | grep -A3 REDACTED | grep -E
  "Location|w="
- Your Highlight on Location 47-48
- Your Note on Location 48
w=17/19 md5(w)=67
- Your Highlight on Location 104-105
- Your Note on Location 105
w=11/15 md5(w)=c6
- Your Highlight on Location 200
- Your Note on Location 200
w=2/3 md5(w)=4b
- Your Highlight on Location 292-293
- Your Note on Location 293
w=12/12 md5(w)=fc

…/Pinaka/reader_home ❱ rev ~/.../q | base64 -d
linux think like a hacker

…/Pinaka/reader_home ❱ 
```

Fig 3 - highlight notes and decoded query

4. Identify the book

Searching amazon.in Kindle for *linux think like a hacker*: red letters, hooded figure, author **Kushagra Dwivedi** (K + 7, D + 6 fits the mask). **Break the Illusion: Getting Started with Linux**, ASIN `B0GHNJYNHH`. reading-list.md also said 'finish Part VI', and the book has six parts.

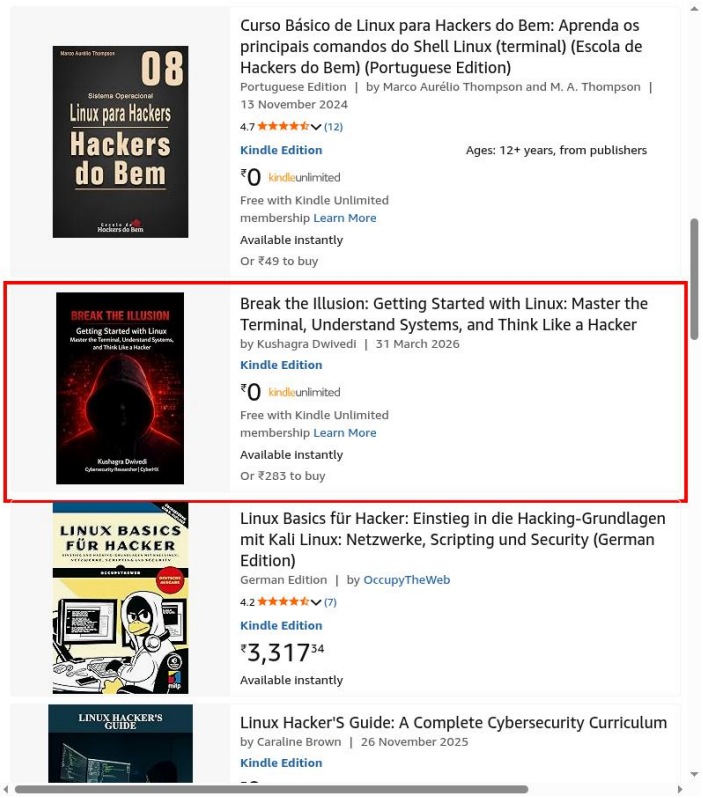


Fig 4 - amazon.in Kindle search result

5. Pull the words from the free sample

Each note gives word position / word count for one highlight. Tokens are counted the way `pick` does (digits and dashes dropped). Each candidate is then checked against its md5 prefix.

Loc 47-48 (Dedication): *The best hackers are not the ones who break the most - they are the ones who understand the most*. 19 words, word 17 = **understand** (md5 67..)

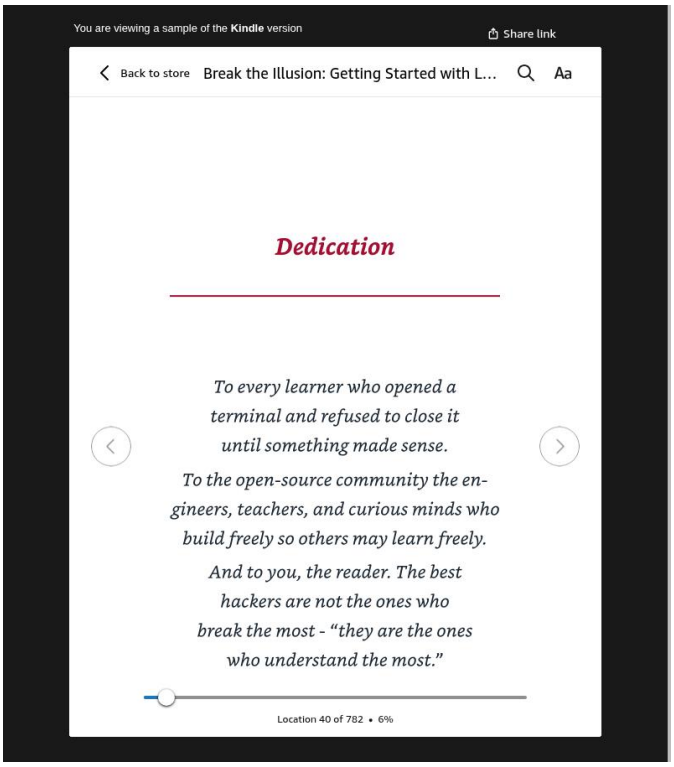


Fig 5 - Kindle sample

Loc 104-105: *Sometimes the secret is already in front of you - just disguised enough to be ignored*. 15 words, word 11 = **disguised** (md5 c6..)

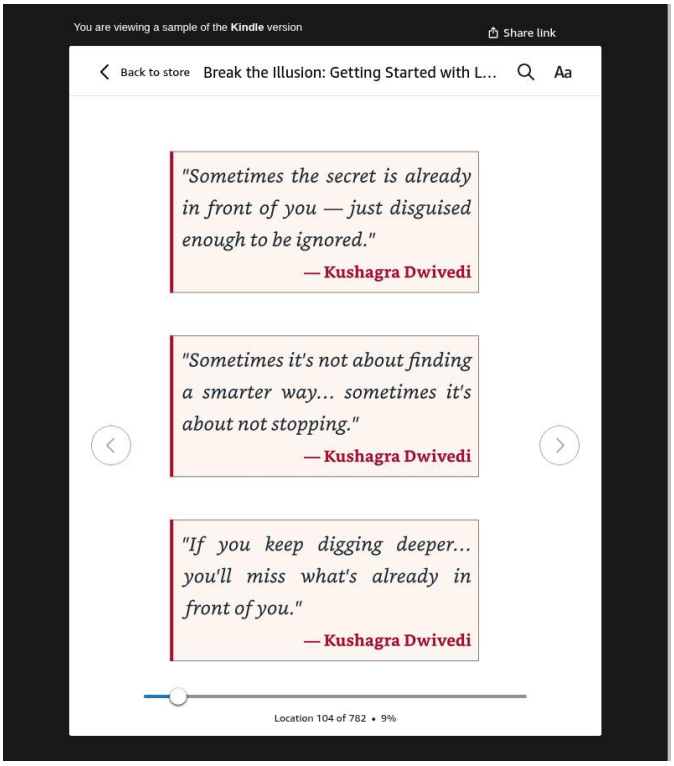


Fig 6 - Kindle sample

Loc 200 (Table of Contents): *Chapter 12 - Permission Management*. Digits dropped leaves 3 words, word 2 = **permission** (md5 4b..)

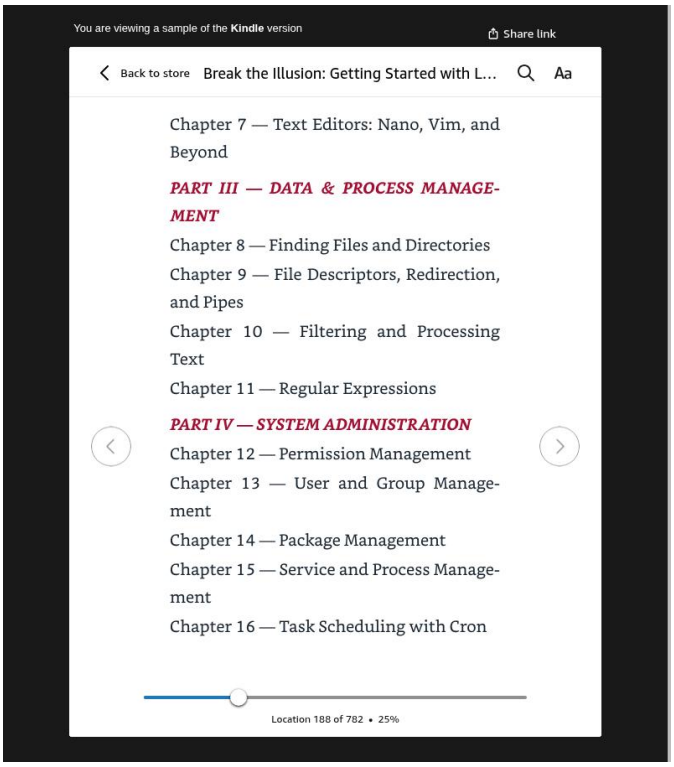


Fig 7 - Kindle sample

Loc 292-293 (Part I intro): *And every line you understand... is one step closer to taking control*. 12 words, word 12 = **control** (md5 fc..)

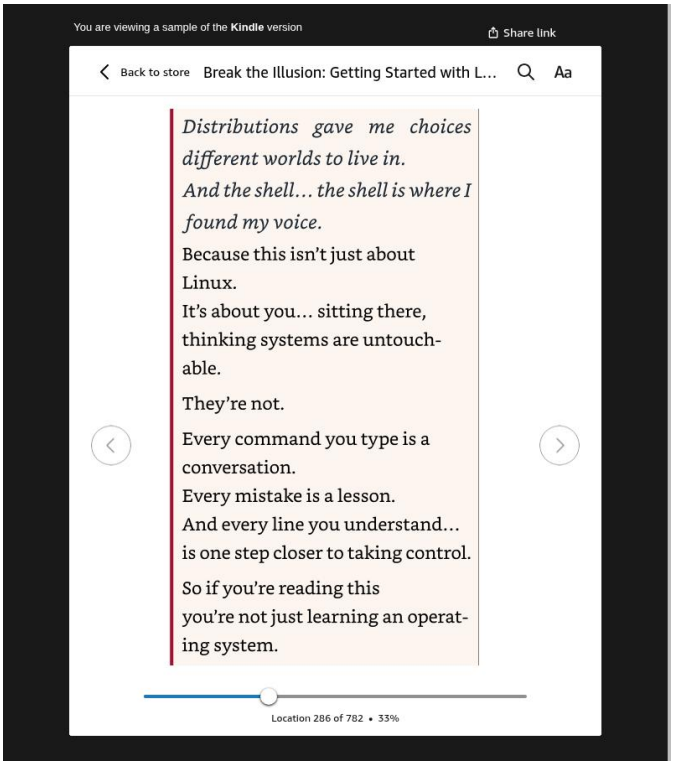


Fig 8 - Kindle sample

6. Verify and decrypt

```
…/Pinaka/reader_home ❱ for w in understand disguised permission control; do echo "$w
  $(printf %s $w | md5sum | cut -c1-2)"; done
understand 67
disguised c6
permission 4b
control fc

…/Pinaka/reader_home ❱ printf '%s' 'understand-disguised-permission-control-b0ghnjynhh' > k

…/Pinaka/reader_home ❱ openssl enc -d -aes-256-cbc -pbkdf2 -iter 250000 -md sha256 \
    -in reader/.vault/flag.enc -pass file:k
CyberHX{y0u_d0nt_l34rn_l1nux_y0u_t4k3_c0ntr0l}

"Sometimes it's not about hacking harder... it's about looking where others don't."
Break the Illusion: Getting Started with Linux - Kushagra Dwivedi

…/Pinaka/reader_home ❱ 
```

Fig 9 - md5 prefixes match, flag decrypted

Solve script

```
tar xzf "reader_home (1).tar.gz"
printf '%s' 'understand-disguised-permission-control-b0ghnjynhh' > k
openssl enc -d -aes-256-cbc -pbkdf2 -iter 250000 -md sha256 \
  -in reader/.vault/flag.enc -pass file:k
```

Flag

CyberHX{y0u_d0nt_l34rn_l1nux_y0u_t4k3_c0ntr0l}